



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

FINAL EXAMINATION DECEMBER 2025

Course : EC3122 (Web Development)
Lecturer : Muhammad Hamizan Johari

Time: 8.00 am – 11.00 am (3 hours)
Date : 13 February 2026

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of 4 printed pages.
(excluding front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1. a. Briefly describe the following components related to the World Wide Web (WWW).
 - i. Web Pages (2 MARKS)
 - ii. Web Browsers (2 MARKS)
 - iii. Hyperlinks (2 MARKS)
- b. Briefly discuss at least **THREE (3)** main website designs used in web development. (6 MARKS)
- c. Briefly explain the following document types. (4 MARKS)
 - i. Transitional document
 - ii. Strict document
- d. Briefly discuss the differences between HTML4 and HTML5 based on these following comparison topics.
 - i. Syntax differences. (3 MARKS)
 - ii. Browser support. (3 MARKS)
 - iii. Future-proofing (3 MARKS)
2. a. Briefly explain each of the following HTML semantic elements.
 - i. <article> tag (2 MARKS)
 - ii. <aside> tag (2 MARKS)
 - iii. <footer> tag (2 MARKS)
 - iv. <header> tag (2 MARKS)
 - v. <nav> tag (2 MARKS)

b.

Item Class	Item Name	Item Description
Phone	iPhone 15	Apple smartphone with A16 Bionic chip, dual-camera system, and 5G support.
Phone	Samsung Galaxy S24	Samsung flagship phone with high-performance processor, AMOLED display, and 5G.
Accessory	USB-C Charger	Fast-charging USB-C wall adapter compatible with most smartphones.
Accessory	Wireless Earbuds	Bluetooth earbuds with charging case and noise reduction for calls.

Write a HTML coding to describe a list of items in a phone store based on the given table with the item class and item name bolded. **(6 MARKS)**

c. Write a HTML coding to create a table based on the following descriptions with a page title and italic each of the table header. The table header must be appropriate based on the given descriptions. **(9 MARKS)**

- i. Kumori, Tokyo City, Japan
- ii. Paul, Manchester City, England
- iii. Abuela, Mexico City, Mexico

3. a. Briefly explain the following types of CSS selector with **ONE (1)** example.

i. Class selector.

(2 MARKS)

ii. Element selector.

(2 MARKS)

iii. ID selector.

(2 MARKS)

b. Briefly discuss the main reasons to apply CSS (Cascading Style Sheet) for developing a website.

(4 MARKS)

c. Given a set of CSS styling requirements: a light gray background color, Arial as the text style, and dark red font color, write the code to apply these CSS styles using the following techniques.

- i. Inline CSS (5 MARKS)
- ii. Internal CSS (5 MARKS)
- iii. External CSS (5 MARKS)

4. a. Rewrite the following switch case statement into an if...else statement. (6 MARKS)

```
function determineHouseColorByNumber(houseNumber) {
  if (houseNumber <= 0) {
    console.log("Invalid house number.");
    return;
  }
```

```
// Convert houseNumber into a group code (1 to 6)
let group;
if (houseNumber <= 20) group = 1;
else if (houseNumber <= 40) group = 2;
else if (houseNumber <= 60) group = 3;
else if (houseNumber <= 80) group = 4;
else if (houseNumber <= 100) group = 5;
else group = 6;
```

```
switch (group) {
  case 1:
    console.log("House color: Red (1-20).");
    break;
  case 2:
    console.log("House color: Blue (21-40).");
    break;
  case 3:
    console.log("House color: Green (41-60).");
    break;
  case 4:
    console.log("House color: Yellow (61-80).");
    break;
  case 5:
    console.log("House color: Purple (81-100).");
    break;
  case 6:
    console.log("House color: Black (101+).");
    break;
  default:
    console.log("Invalid house number.");
}
```

- b. Given the nested do...while loop and the array of fruits:

(6 MARKS)

```
const nepaleseCuisine = [
  "Momo",
  "Dal Bhat",
  "Sel Roti",
  "Gundruk",
  "Thukpa",
  "Chatamari"
];

let i = 0;

do {
  let j = i + 1;

  // only run inner loop if there is at least one valid pair
  if (j < nepaleseCuisine.length) {
    do {
      console.log(`${nepaleseCuisine[i]} and
      ${nepaleseCuisine[j]}`);
      j++;
    } while (j < nepaleseCuisine.length);
  }

  i++;
} while (i < nepaleseCuisine.length - 1);
```

Interpret the output of the nested for loop example.

- c. Write a JavaScript function to multiply two numbers inputted by users. **(8 MARKS)**
- d. Write a JavaScript function in a HTML file to display a "Hello, World!" message when the user clicks a "View" button. **(5 MARKS)**

-END OF QUESTION PAPER-